IMPaCT-Data Biomedical Cloud

An initial iteration for a federated virtual computing environment in the context of Precision Medicine in Spain

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I. BACKGROUND

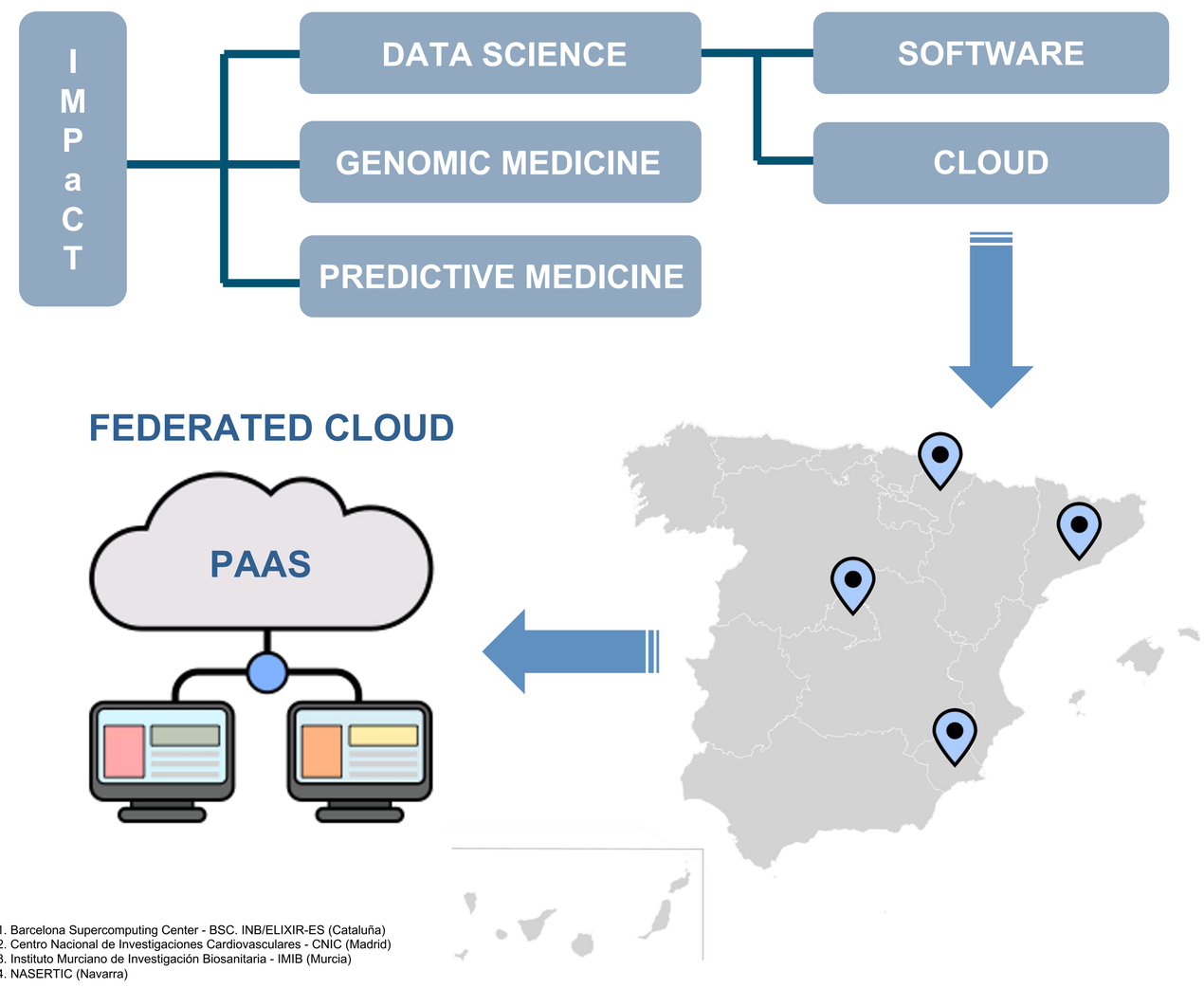
The Spanish Precision Medicine Infrastructure associated with Science and Technology (**IMPaCT**), aims to impulse **precision medicine** within the Spanish National Health System. **IMPaCT-Data**, the Data Science pillar of this infrastructure, has been tasked to design, deploy and extend a **Biomedical Cloud** taking as reference similar efforts across Europe.

II. OBJECTIVE

The establishment of IMPaCT-Data **Biomedical Cloud** aim to provide a **scalable** and **flexible** analysis **environment**, enabling the **integration**, **management** and **analysis** of sensitive and non-sensitive data, e.g. richly described genomics, structured clinical, and medical imaging data, made available within IMPaCT for research purposes.

III. STRUCTURE

The infrastructure is designed as a **federated cloud system,** which will include a *Life Sciences Login*-compliant **authentication system**, a data infrastructure to manage both **public and sensitive data**, and a **distributed** set of **computational resources**, which will be assigned depending on the data that should be mobilized, and the needs of the desired analysis.



USERS

AUTHENTICATION

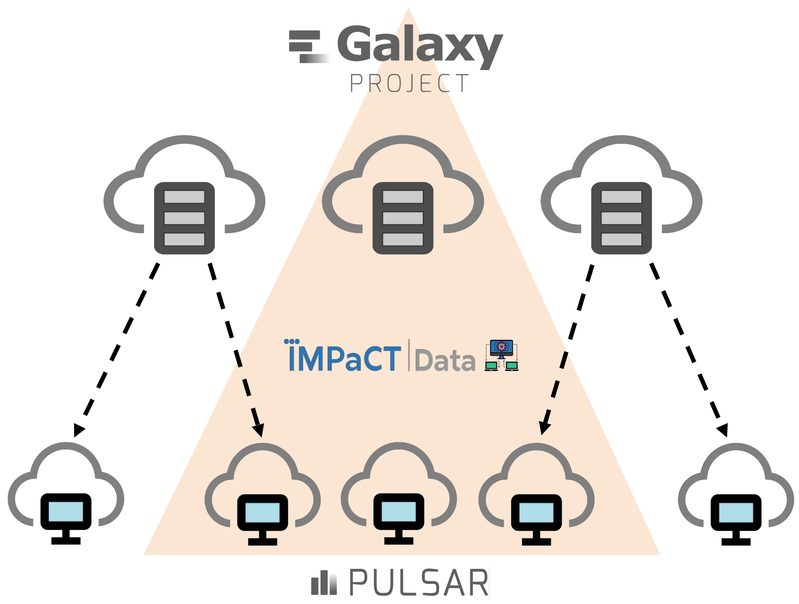
- **FEDERATED AUTHENTICATION:** SEVERAL IdPs AND A CENTRAL NODE
- SINGLE SET OF CREDENTIALS FOR MULTIPLE SYSTEMS (**SSO**)
- AUTHENTICATION: OpenID CONNECT **KEYCLOAK**
- AUTHORIZATION: **GA4GH** PASSPORT

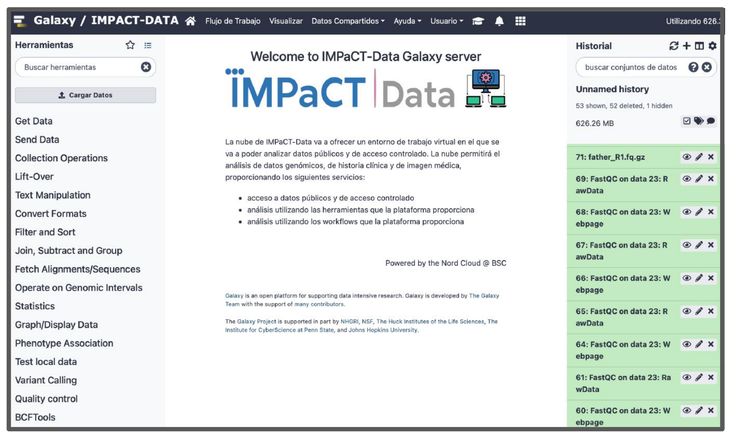
PUBLIC DATA: CAN BE ACCESSED BY ANYONE WITH A **VALID IMPaCT ACCOUNT (GALAXY, PULSAR, CVMFS)**

PROTECTED DATA: ONLY **APPROVED USERS** WILL BE ABLE TO INTERACT WITH **ACCESS CONTROLLED** DATA (**FEDERATED EGA**-LIKE)

IV. ARCHITECTURE

- Structure based on the connection of **one main node** (IMPaCT) and **three pulsar workers** (BSC, IMIB, CNIC)
- **Parallel jobs** proceeding from the execution of a **single tool** within the IMPaCT environment can be executed in **different cores.**
- Pulsar nodes can be connected to **several main** Galaxy servers, even if they are outside of the IMPaCT network and receive jobs from both of them **simultaneously**.





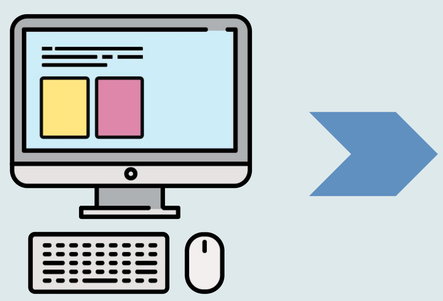
- Access **sample data** thanks to CVMFS, load **local** files or access information from a customizable **cloud space**.
- Users can also perform their analysis by using IMPaCT-Data **pre-installed** tools and workflows or **upload** their own.

V. THE FUTURE

1. Implementation of **data encryption protocols** for **protected** data handling.

2. Standarization of ansible recipes for **easier enviromnent reproductibility**.

3. Inclusion of **new platforms** for data **analysis** and **processing.**



bio.tools
BioContainers
WorkflowHub
OpenEBench
EGA

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